

Real DX Risk Geometry Bridge

Phase 15 moves the 4-asset geometry program from synthetic prompts to real DX observation prompts. The question is not whether risk changes the final action, but whether changing *Asset Risk Preference* from 1 to 5 deforms the *row-local* market geometry carried by the selected asset rows.

DATE	REAL DATA	LADDER	OBJECT
22 March 2026	30 matched HQ observations	risk_1 .. risk_5, anchored on risk_3	row-local 4-asset geometry

MAIN READ

The synthetic risk-deformation story does *not* transfer to row-local geometry on matched real DX prompts. In the real bridge, the selected 4-asset row geometry stays effectively fixed across risk_1 .. risk_5.

Cross-example coordinate probes are only moderate ($0.576\ R^2$ for the best x-axis read; $0.466\ R^2$ for the best y-axis read), so the real hand-built coordinate frame is not decoded nearly as cleanly as in synthetic phases. But the more important result is stronger: the decoded row-local geometry does not move with risk. Realignment margins are slightly negative at every context, and the decoded geometry-step norm is exactly 0.0 for every adjacent real risk pair. That implies the row-local market frame is stable and the risk setting is likely being applied somewhere beyond the individual market-row states.

BASE DATA	BEST X TRANSFER	BEST REALIGNMENT	RISK-STEP DELTA
30 / 150	0.576 R^2	-0.0059	0.0
base examples / prompts	row_mean @ L21	still negative, row_eos @ L4	decoded geometry-step norm

Why This Bridge Exists

Phases 9-14 built a synthetic story around a shared 4-asset market frame:

- the model preserved a reusable coordinate system over multiple assets
- synthetic risk behaved like a structured ladder on top of that frame
- synthetic portfolio and affordance produced different families of deformations

Phase 15 asks the obvious external-validation question:

- if we take *real DX observation prompts*
- keep the market snapshot fixed

- and change only the risk preference setting

do the same row-local geometry effects appear?

This is a bridge test, not a restatement of the synthetic results. If the bridge fails, that is scientifically useful because it tells us where the synthetic abstraction stopped matching the real system.

Experimental Design

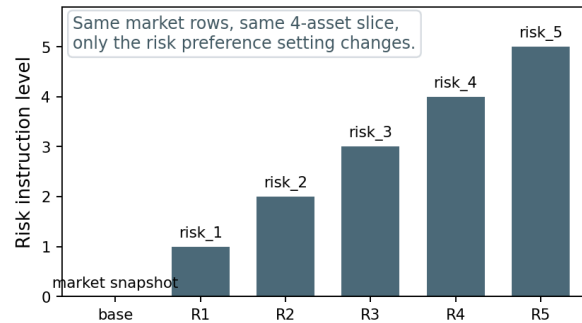
What Data Was Used

1. 30 matched real base examples
2. 150 total prompts across risk_1..risk_5
3. 5 roster families x 6 examples each
4. 4-asset slice chosen inside each larger real roster

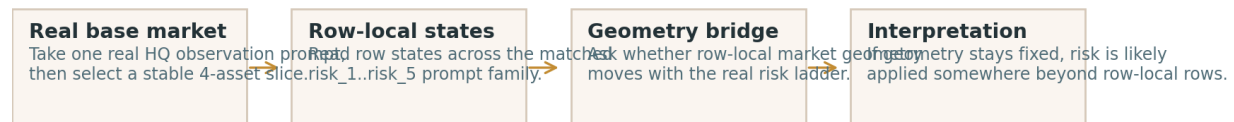
Preference is edited across variants

Base examples: 30
Prompts: 150
Contexts: 5
Rosters: 5

What The Real Risk Ladder Means



What Is Being Tested



The real bridge dataset recipe. Each base example is a real HQ observation prompt. A stable 4-asset slice is selected inside that larger roster, then the prompt is rerendered at risk_1 .. risk_5 with the market snapshot held fixed.

The dataset contains:

- 30 matched real base examples
- 150 total prompts
- 5 real roster families, each contributing 6 examples
- 4 selected assets per base example, chosen from a larger real roster

Important constraint:

- there is no natural supply of HQ prompts with exactly 4 assets
- so this bridge works with a *4-asset slice* inside larger real rosters
- the slice is kept fixed across the whole risk ladder for that example

In plain English:

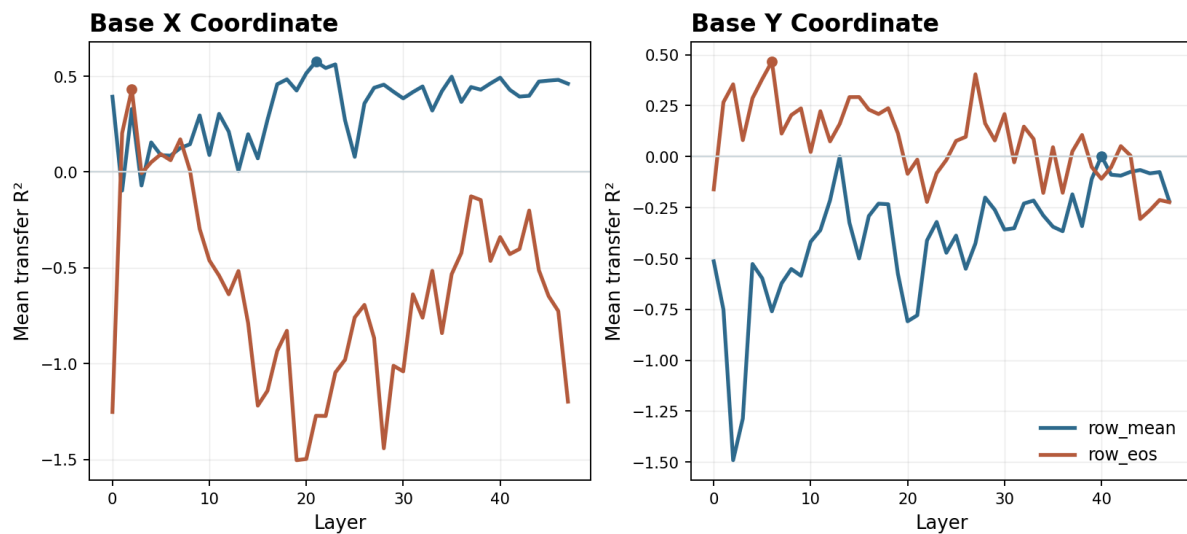
- same market rows
- same 4 selected assets
- same row order and symbols inside the selected slice
- only the risk preference setting changes

Plain-Language Definitions

- row-local geometry the multi-asset shape recovered only from the selected asset rows, not from later policy sections or the final action token
- risk ladder the ordered family `risk_1`, `risk_2`, `risk_3`, `risk_4`, `risk_5` for the same underlying prompt
- coordinate transfer how well a probe trained to recover a real coordinate axis from one risk context generalizes to other risk contexts
- realignment margin how much closer the recovered geometry is to the risk-adjusted score geometry than to the unchanged base geometry
- risk-adjusted score geometry the asset layout implied if the same market is rescored under a more conservative or more aggressive risk preference
- bridge a synthetic-to-real validation step, not a new synthetic phase

Coordinate Transfer

Cross-Example Coordinate Transfer Across The Real Risk Ladder



Mean coordinate-transfer curves by layer, averaged over the real risk ladder. The x-axis is model layer; the y-axis is transfer R^2 .

The coordinate-transfer chart says two things at once:

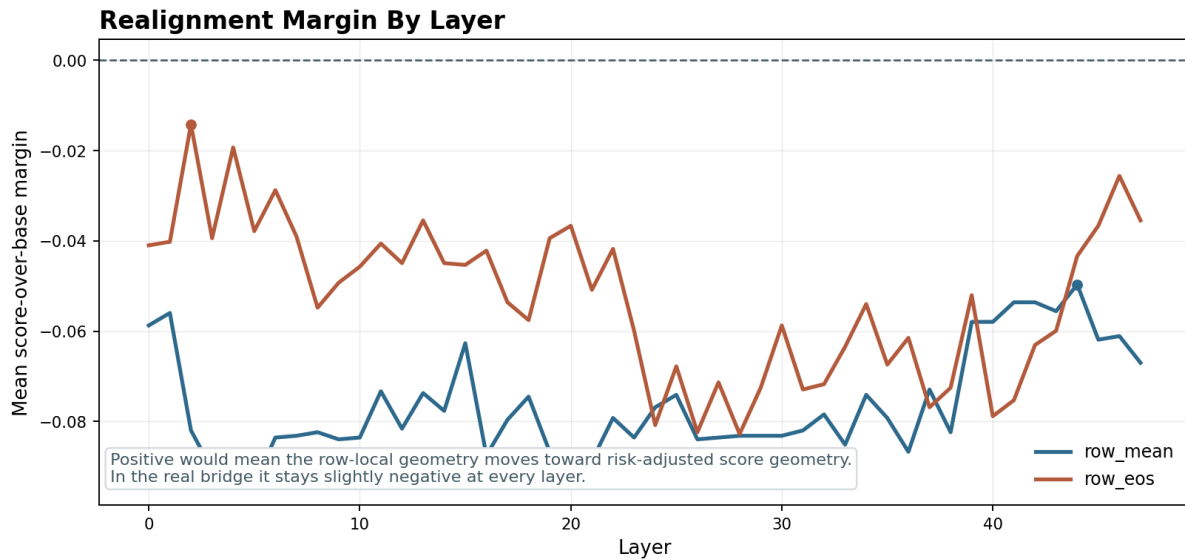
- the real row states do carry some stable multi-asset structure
- but the real hand-built coordinate frame is much noisier than the synthetic one

The best reads are:

- `base_x`: 0.576 R^2 from `row_mean` @ L21
- `base_y`: 0.466 R^2 from `row_eos` @ L6

That is enough to say the real 4-asset object is not random. But it is not enough to claim a crisp low-dimensional real market frame in the same sense as the synthetic phases. So the stronger evidence in this report does *not* come from cross-example coordinate quality. It comes from what happens *across the risk ladder within the same example*.

Realignment Margin



If row-local geometry were moving toward risk-adjusted score geometry, the margin would turn positive. Instead it stays slightly negative.

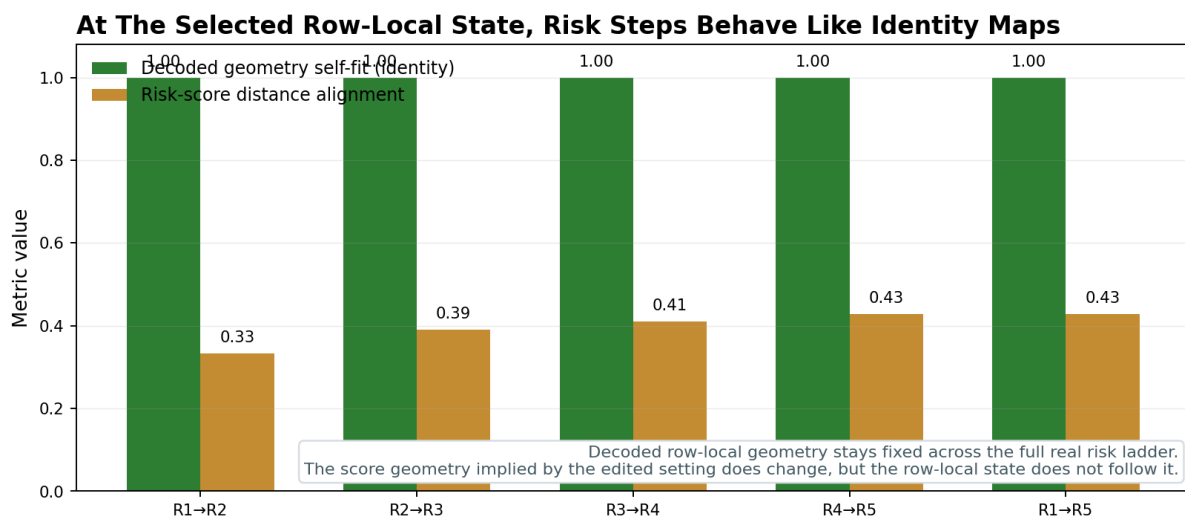
This is the first strong bridge result.

Across all five real risk contexts:

- the best realignment margin is still negative: -0.0059
- there is no layer where the row-local geometry clearly shifts toward risk-adjusted score geometry
- row_eos gets closest, but it still does not cross into positive territory

That means the real risk preference edit is *not visibly deforming the selected row-local geometry in the expected direction*.

Identity Across Real Risk Steps



At the selected row-local state, decoded geometry is unchanged across adjacent real risk steps, while the score geometry implied by the prompt does change.

This is the strongest result in the bridge.

At the selected row-local state (row_eos @ L2):

- each adjacent risk step is fit perfectly by the identity map (coord_r2_mean = 1.0)
- the decoded geometry-step norm is 0.0 for every pair
- but the risk-adjusted score geometry would prefer nontrivial movement (0.33 .. 0.43 mean distance-Spearman)

So the bridge does *not* say “risk has no effect in the real system.”

It says something narrower and more useful:

- risk does not show up as a deformation of the selected row-local 4-asset geometry

That strongly suggests the risk setting is applied:

- later than the market-row states we are reading
- or in a different representation than the row-local market geometry

Interpretation

The cleanest reading of Phase 15 is:

1. Synthetic phases were real and useful, but they were still synthetic.
2. Real DX row-local geometry seems to preserve the market slice across the risk ladder.
3. The risk instruction is therefore likely being integrated somewhere beyond those row-local states.

This is a meaningful narrowing of the problem:

- if we want the real risk effect, more row-local market probing will probably not be enough
- the next place to look is *context integration over the market frame*, not the market rows in isolation

In other words, the bridge did not fail because it found nothing. It failed in a specific way:

- the selected row-local market geometry is too stable
- so risk is probably applied downstream of it

What To Do Next

The next move should be narrower than another large synthetic expansion.

Best next step:

- run the same bridge lens on *context-integration states*, not only row-local row states

Concretely:

- keep the same matched real risk ladder
- keep the same selected 4-asset slices
- move the representation object to section-level or later aggregate states

Examples:

- portfolio or settings section EOS states
- later token positions after the model has seen both the market and the user settings
- pairwise or set-level comparisons formed after the full market block, not inside individual rows

That would directly test the current hypothesis:

- the real risk effect exists
- but it is applied after the shared market slice has already been encoded

How To Read The Layerwise Charts

The layerwise charts in this report all follow the same pattern:

- x-axis = layer
- y-axis = average metric value
- a peak means “this read is strongest here”

The two main y-axis metrics are:

- transfer R^2 how well a coordinate probe recovers a real axis from held-out rows
- realignment margin whether recovered geometry is closer to risk-adjusted score geometry or to unchanged base geometry

Interpretation guide:

- high transfer + positive margin would support real row-local deformation
- high transfer + negative margin means the representation is stable, but not moving with the edited context
- flat zero deformation means the row-local geometry is effectively unchanged across the ladder

That third case is what dominates this report.